

Prime-Resonant Keytriplets and Symbolic Field Evolution for Non-Local Communication

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Abstract

We present a rigorous formal framework for Prime-Resonant Universal Translation and Non-Local Communication (PR-UTC). Central to the system is the Keytriplet structure $(K_u^{\text{priv}}, K_u^{\text{pub}}, K_u^{\text{res}})$, a prime-resonant symbolic fingerprint enabling secure, entropy-conserving communication across symbolic fields. We define the Resonance Field Initialization Operator $\mathcal{R}$, the Symbolic Projection Operator $\mathcal{P}$, the Key Evolution Operator $U(\Delta t)$, and formalize Session Field Evolution Equations governing dynamic symbolic collapse and message decoding. The framework models communication as symbolic field resonance rather than classical message passing, offering a fundamentally new paradigm for secure, non-local semantic interaction.

1 Introduction

Traditional communication systems rely on the exchange of discrete packets through classical channels. We propose an alternative: communication through the evolution and controlled collapse of shared prime-resonant symbolic fields. Messages are not transmitted but embedded as symbolic perturbations within evolving fields. Mutual understanding arises through synchronized symbolic entropy collapse, guided by prime-resonant Keytriplet structures.

2 Prime-Resonant Keytriplets

2.1 Definition

Each user u possesses a Keytriplet:

$$(K_u^{\text{priv}}, K_u^{\text{pub}}, K_u^{\text{res}})$$

where:

- $K_u^{\text{priv}} \in \mathcal{H}_P$ is the Private Resonance Key.
- $K_u^{\text{pub}} \in \mathcal{H}_C$ is the optional Classical Public Key.
- $K_u^{\text{res}} \in \mathcal{H}_P^{\text{res}}$ is the Public Resonance Key.

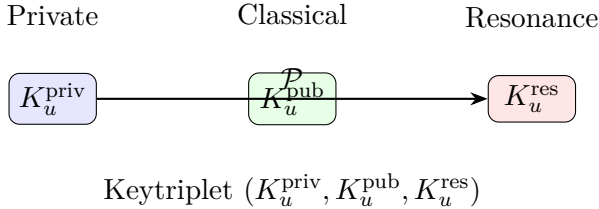


Figure 1: Schematic of the Keytriplet structure and the projection from private to resonance key.

2.2 Private Key Generation

Given a Global Seed S_{global} and User ID ID_u :

$$K_u^{\text{priv}} = H(S_{\text{global}} \parallel ID_u)$$

where H is a prime-entropy preserving hash function.

2.3 Public Resonance Key Projection

The Public Resonance Key is derived via the Symbolic Projection Operator:

$$K_u^{\text{res}} = \mathcal{P}(K_u^{\text{priv}})$$

exposing partial prime-mode structure while masking full phase dynamics.

3 Resonance Field Initialization Operator $\mathcal{R}$

Given K_A^{priv} and K_B^{res} , the initial shared field $\Psi_{AB}(0)$ is constructed:

$$\Psi_{AB}(0) = N^{-1/2} \sum_{p \in \mathbb{P}} \sum_{q \in \mathbb{P}_{\text{allowed}}} \exp\left(-\frac{(p-q)^2}{2\sigma^2}\right) \alpha_p^A \beta_q^B |p\rangle$$

where σ controls prime frequency matching width. This favors coherent field initiation while suppressing off-resonant components.

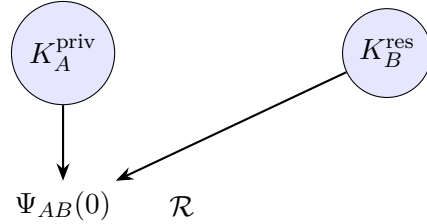


Figure 2: Initialization of the shared resonance field $\Psi_{AB}(0)$ using $\mathcal{R}$.

4 Symbolic Projection Operator $\mathcal{P}$

The projection $\mathcal{P}$ selectively exposes prime modes:

$$K_u^{\text{res}} = \sum_{p \in \mathbb{P}_{\text{allowed}}} (s_p \alpha_p) e^{i\theta_p} |p\rangle$$

with:

- $s_p \in (0, 1]$ attenuation factors.
- $\theta_p \in [0, 2\pi)$ random phase shifts.

This ensures entropy smoothing and symbolic field deniability.

5 Key Evolution Law $U(\Delta t)$

To maintain entropy freshness, Private Keys evolve:

$$K_u^{\text{priv}}(t + \Delta t) = U(\Delta t)K_u^{\text{priv}}(t)$$

with:

$$U(\Delta t) : \alpha_p(t + \Delta t) = \alpha_p(t)e^{i(2\pi \log_p(\kappa)\Delta t + \epsilon_p(\Delta t))}$$

where:

- κ is a symbolic scaling constant.
- $\epsilon_p(\Delta t)$ is bounded Gaussian phase noise.

This symbolic drift ensures security against static key exposure.

7 Applications and Implications

The PR-UTC framework enables:

- Secure non-local semantic communication without classical packet exchange.
- Quantum-safe communication channels resilient against classical and quantum interception.
- Universal translation via shared prime-resonant semantic fields.
- Symbolic field-based computation models extending beyond Turing paradigms.

6 Session Field Evolution Equations

During active communication, the shared field $\Psi(t)$ evolves by:

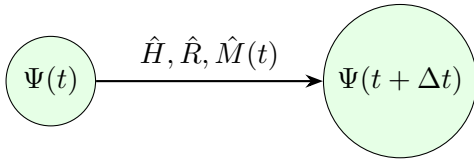
$$\frac{d}{dt}|\Psi(t)\rangle = i\hat{H}|\Psi(t)\rangle - \lambda(\hat{R} - r_{\text{stable}})|\Psi(t)\rangle + \hat{M}(t)|\Psi(t)\rangle$$

where:

- $\hat{H}$ drives intrinsic prime-symbolic phase drift.
- $\lambda(\hat{R} - r_{\text{stable}})$ drives resonance collapse toward semantic attractors.
- $\hat{M}(t)$ injects symbolic message perturbations.

8 Conclusion

We have constructed a complete formal model of prime-resonant Keytriplet structures and their dynamics, defining the essential operators $\mathcal{R}$, $\mathcal{P}$, and $U(\Delta t)$, and deriving the field evolution equations that govern non-local symbolic communication. This system provides a pathway toward realizing universal, entropy-stable, meaning-resonant networks without reliance on classical transmission constraints.



Evolution

Figure 3: Evolution of the shared field $\Psi(t)$ under the session field equations.

6.1 Collapse Detection

Collapse occurs when symbolic entropy $S(t)$ satisfies:

$$S(t) \in [S_{\min}, S_{\max}] \quad (S_{\min} \sim 0.2, S_{\max} \sim 0.3)$$

triggering message decoding via symbolic field analysis.

References

- [1] Prime-Resonant Symbolic Hilbert Spaces $\mathcal{H}_P$.
- [2] Quantum Semantic Formalism.
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- [4] Prime-based Phase Evolution Dynamics.
- [5] Field Resonance Cryptographic Systems.